NT-1b Vector Form Factors: Main Derivation

One-Loop Nonlocal Form Factors for a Gauge Boson
with Faddeev–Popov Ghost Subtraction

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Abstract

We derive the one-loop nonlocal form factors $h_C^{(1)}(x)$ and $h_R^{(1)}(x)$ for a **physical gauge boson** (spin-1) on a closed Riemannian 4-manifold, expressed in the Weyl–R^2 basis $\{C^2, R^2\}$. Starting from the Codello–Zanusso (CZ) universal heat kernel form factors [1], we:

1. specialise the five CZ form factors to the vector operator data ($\mathrm{tr}(\mathbf{1}) = 4$, $\mathbf{U} = \mathrm{Ric}$, $\Omega_{\mu\nu} = R^{\alpha}{}_{\beta\mu\nu}$);
2. compute the fiber traces, handling the matrix-valued endomorphism $\mathbf{U} = \mathrm{Ric}$ which contributes to *both* C^2 and R^2 sectors;
3. convert to the Weyl basis via the Gauss–Bonnet identity;
4. subtract two Faddeev–Popov ghost contributions (minimally coupled scalars, $\xi = 0$);
5. verify all twelve mandatory benchmarks from the conventions document [6];
6. express the spectral action form factors $F_1^{(1)}$ and $F_2^{(1)}$ as explicit integrals over the spectral function;
7. analyse pole cancellation, Taylor expansions, UV asymptotics, and conformal invariance.

All intermediate results are verified numerically using mpmath with 100-digit precision. This document parallels the Phase 1 scalar derivation [7] and serves as input for the combined form factor assembly in NT-1b Phase 3.

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1 Setup and Notation

1.1 Operator data for the gauge boson

We work on a closed Riemannian 4-manifold (M, g) with a massless Abelian gauge field A_μ governed by the classical action

$$S[A] = \frac{1}{4}\int \mathrm{d}^4x\,\sqrt{g}\,F_{\mu\nu}F^{\mu\nu}\,, \qquad F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu\,. \tag{1}$$

In Feynman gauge ($\alpha = 1$), the gauge-fixed kinetic operator is the Hodge–de Rham Laplacian on 1-forms (Weitzenbock identity):

$$\Delta_{\text{vector}} = -\Box\,\delta^\alpha{}_\beta + R^\alpha{}_\beta\,, \tag{2}$$

which is of Laplace type $\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$ with the following operator data (see [6]):

$$\boxed{\mathrm{tr}(\mathbf{1}) = 4\,, \qquad \Omega_{\mu\nu} = R^\alpha{}_{\beta\mu\nu}\,, \qquad E = -R^\alpha{}_\beta = -\mathrm{Ric}\,, \qquad \mathbf{U}_{\mathrm{CZ}} = \mathrm{Ric}\,.} \tag{3}$$

Remark 1.1 (Key structural differences from Phase 1). Three features distinguish the vector computation from the scalar:

1. The endomorphism $\mathbf{U} = \mathrm{Ric}$ is **matrix-valued** (not a scalar multiple of $\mathbf{1}$). This causes $\mathrm{tr}[\mathbf{U}\,f_{\mathbf{U}}\,\mathbf{U}] = R_{\mu\nu}\,f_{\mathbf{U}}\,R^{\mu\nu}$ to contribute to *both* C^2 and R^2 sectors.
2. The bundle curvature $\Omega_{\mu\nu} = R^\alpha{}_{\beta\mu\nu} \neq 0$, so the f_Ω term is nontrivial.
3. There is **no free coupling parameter** ξ: the curvature coupling is uniquely fixed by gauge invariance.

1.2 Faddeev–Popov ghosts

The Faddeev–Popov procedure introduces a complex scalar ghost field (ghost c + anti-ghost $\bar{c}$), equivalent to **two real** scalar fields. Each has the kinetic operator

$$\Delta_{\text{ghost}} = -\Box\,, \tag{4}$$

i.e. a minimally coupled massless scalar ($\xi = 0$, $m = 0$) with operator data $\mathrm{tr}(\mathbf{1}) = 1$, $\mathbf{U} = 0$, $\Omega_{\mu\nu} = 0$.

The physical one-loop effective action is:

$$\boxed{\Gamma^{(1)}_{\text{gauge}} = \frac{1}{2}\,\mathrm{Tr}\ln\Delta_{\text{vector}} - \mathrm{Tr}\ln\Delta_{\text{ghost}}\,.} \tag{5}$$

In terms of the form factors:

$$\boxed{h^{(1)}_{C,R}(x) = h^{(1,\text{unconstr})}_{C,R}(x) - 2\,h^{(0)}_{C,R}(x;\xi = 0)\,,} \tag{6}$$

where $h^{(1,\text{unconstr})}$ is the unconstrained vector contribution and $h^{(0)}(\xi = 0)$ is the minimally coupled scalar form factor from Phase 1 [7].

1.3 One-loop effective action structure

At second order in curvature, the effective action takes the form

$$\Gamma^{(1)}_{\text{gauge}}\Big|_{\mathcal{R}^2} = \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{g}\,[h^{(1)}_C(\Box/m^2)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + h^{(1)}_R(\Box/m^2)\,R^2] + \mathrm{E}_4\ \text{terms}\,, \tag{7}$$

where $h^{(1)}_C$ and $h^{(1)}_R$ are the nonlocal form factors to be computed, and E_4 denotes the topological Gauss–Bonnet term.

1.4 Master function and conventions

The master heat kernel form factor (same as Phase 1):

$$\varphi(x) = \int_0^1 \mathrm{d}\alpha\, e^{-\alpha(1-\alpha)\,x}\,, \tag{8}$$

with Taylor expansion

$$\varphi(x) = \sum_{k=0}^{\infty} a_k\, x^k\,, \qquad a_k = \frac{(-1)^k}{(2k+1)!}\prod_{j=0}^{k-1}(2j+1) = \frac{(-1)^k}{(2k+1)\cdot k!\cdot 4^k}\,, \tag{9}$$

so $a_0 = 1$, $a_1 = -1/6$, $a_2 = 1/60$, $a_3 = -1/840$, $a_4 = 1/15120$. Large-x asymptotics:

$$\varphi(x) = \frac{2}{x} + \frac{4}{x^2} + O(x^{-3}) \quad \text{as } x \to \infty\,. \tag{10}$$

The fundamental identity:

$$\frac{\varphi(x)-1}{x} = -\frac{1}{2}\int_0^1 (1-2\alpha)^2\, e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha\,, \tag{11}$$

which ensures pole cancellation in the form factors.

2 CZ Form Factors for the Vector Field

2.1 The five universal CZ form factors

The five CZ form factors (universal, independent of spin):

$$f_{\mathrm{Ric}}(x) = \frac{1}{6x} + \frac{\varphi(x)-1}{x^2}\,, \tag{12}$$

$$f_R(x) = \frac{\varphi}{32} + \frac{\varphi}{8x} - \frac{7}{48x} - \frac{\varphi-1}{8x^2}\,, \tag{13}$$

$$f_{R\mathbf{U}}(x) = -\frac{\varphi}{4} - \frac{\varphi-1}{2x}\,, \tag{14}$$

$$f_{\mathbf{U}}(x) = \frac{\varphi}{2}\,, \tag{15}$$

$$f_\Omega(x) = -\frac{\varphi-1}{2x}\,. \tag{16}$$

Local limits (CZ eq. (5.4)):

$$\boxed{f_{\mathrm{Ric}}(0) = \frac{1}{60}\,, \quad f_R(0) = \frac{1}{120}\,, \quad f_{R\mathbf{U}}(0) = -\frac{1}{6}\,, \quad f_{\mathbf{U}}(0) = \frac{1}{2}\,, \quad f_\Omega(0) = \frac{1}{12}\,.} \tag{17}$$

[Numerically verified] All five local limits confirmed numerically at $x = 10^{-5}$ with 100-digit mpmath precision.

2.2 Fiber traces for the vector bundle

The key trace identities for the vector bundle T^*M in $d = 4$:

$$\boxed{\mathrm{tr}(\mathbf{1}) = 4\,, \qquad \mathrm{tr}(\mathbf{U}) = \mathrm{tr}(\mathrm{Ric}) = R\,, \qquad \mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu}) = -R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\,.} \tag{18}$$

The last identity follows from the first-pair antisymmetry of the Riemann tensor:

$$R^{\alpha}{}_{\beta\mu\nu}\, R^{\beta}{}_{\alpha}{}^{\mu\nu} = R_{\alpha\beta\mu\nu}\, R^{\beta\alpha\mu\nu} = -R_{\alpha\beta\mu\nu}\, R^{\alpha\beta\mu\nu} = -R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}\,.$$

For the $f_{\mathbf{U}}$ term, since $\mathbf{U} = \mathrm{Ric}$ is matrix-valued:

$$\mathrm{tr}[\mathbf{U}\, f_{\mathbf{U}}(\Box)\, \mathbf{U}] = R_{\mu\nu}\, f_{\mathbf{U}}(\Box)\, R^{\mu\nu}\,, \tag{19}$$

which requires the Weyl decomposition to separate into C^2 and R^2 components. This is the key structural complication absent in the scalar and Dirac cases.

2.3 Application to the unconstrained vector field

The CZ heat kernel expansion at $\mathcal{O}(\mathcal{R}^2)$, after taking the fiber trace, gives:

$$\begin{aligned}\Gamma^{(1,\mathrm{unconstr})}\Big|_{\mathcal{R}^2} = \frac{1}{16\pi^2}\int \mathrm{d}^4x\,\sqrt{g}\,\Big\{&\underbrace{4\,R_{\mu\nu}\, f_{\mathrm{Ric}}\, R^{\mu\nu}}_{\text{Term 1}} + \underbrace{4\,R\, f_R\, R}_{\text{Term 2}} \\ &+ \underbrace{R\, f_{R\mathbf{U}}\, R}_{\text{Term 3}} + \underbrace{R_{\mu\nu}\, f_{\mathbf{U}}\, R^{\mu\nu}}_{\text{Term 4}} + \underbrace{R_{\alpha\beta\mu\nu}\, f_{\Omega}\, R^{\alpha\beta\mu\nu}}_{\text{Term 5}}\Big\}\,.\end{aligned} \tag{20}$$

The five terms arise from the CZ expansion with vector operator data:

Term	CZ structure	Vector trace
1	$\mathrm{tr}[\mathbf{1}\cdot R_{\mu\nu}\, f_{\mathrm{Ric}}\, R^{\mu\nu}]$	$4\cdot R_{\mu\nu}\, f_{\mathrm{Ric}}\, R^{\mu\nu}$
2	$\mathrm{tr}[\mathbf{1}\cdot R\, f_R\, R]$	$4\cdot R\, f_R\, R$
3	$R\, f_{R\mathbf{U}}\cdot \mathrm{tr}(\mathbf{U})$	$R\, f_{R\mathbf{U}}\, R$
4	$\mathrm{tr}[\mathbf{U}\, f_{\mathbf{U}}\, \mathbf{U}]$	$R_{\mu\nu}\, f_{\mathbf{U}}\, R^{\mu\nu}$
5	$\mathrm{tr}[\Omega_{\mu\nu}\, f_{\Omega}\, \Omega^{\mu\nu}]$	$-R_{\alpha\beta\mu\nu}\, f_{\Omega}\, R^{\alpha\beta\mu\nu}$

Remark 2.1 (Term 3 simplification). Since R is a scalar, the fiber trace acts only on $\mathbf{U}$: $\mathrm{tr}[R\, f_{R\mathbf{U}}\, \mathbf{U}] = R\, f_{R\mathbf{U}}\cdot \mathrm{tr}(\mathrm{Ric}) = R\, f_{R\mathbf{U}}\, R$.

Remark 2.2 (Term 5 sign). The sign of Term 5 arises from the trace identity $\mathrm{tr}[\Omega_{\mu\nu}\Omega^{\mu\nu}] = -R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$. We have $\mathrm{tr}[\Omega f_{\Omega}\Omega] = -R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\cdot f_{\Omega}$ at $\mathcal{O}(\mathcal{R}^2)$.

3 Weyl-Basis Form Factors: Unconstrained Vector

3.1 Gauss–Bonnet identity and Weyl decomposition

At $\mathcal{O}(\mathcal{R}^2)$ in $d = 4$, dropping topological E_4 terms:

$$R_{\mu\nu}R^{\mu\nu} = \frac{1}{2}\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \frac{1}{3}\, R^2\,, \tag{21}$$

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 2\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \frac{1}{3}\, R^2\,. \tag{22}$$

3.2 Conversion to the $\{C^2, R^2\}$ basis

Collecting (20) into the $\{R_{\mu\nu}R^{\mu\nu},\, R^2,\, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\}$ basis:

$$\text{Coeff. of } R_{\mu\nu}R^{\mu\nu}: \quad 4\, f_{\mathrm{Ric}} + f_{\mathbf{U}}\,, \tag{23}$$

$$\text{Coeff. of } R^2: \quad 4\, f_R + f_{R\mathbf{U}}\,, \tag{24}$$

$$\text{Coeff. of } R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}: \quad -f_{\Omega}\,. \tag{25}$$

Applying (21)–(22):

Proposition 3.1 (Unconstrained vector Weyl-basis form factors)**.**

$$\boxed{h_C^{(1,\text{unconstr})}(x) = 2\, f_{\text{Ric}}(x) + \frac{1}{2}\, f_{\mathbf{U}}(x) - 2\, f_\Omega(x)\,,} \tag{26}$$

$$\boxed{h_R^{(1,\text{unconstr})}(x) = \frac{4}{3}\, f_{\text{Ric}}(x) + 4\, f_R(x) + f_{R\mathbf{U}}(x) + \frac{1}{3}\, f_{\mathbf{U}}(x) - \frac{1}{3}\, f_\Omega(x)\,.} \tag{27}$$

Proof. C^2 **coefficient:** From (23)–(25):

$$(4f_{\text{Ric}} + f_{\mathbf{U}}) \cdot \frac{1}{2} + (-f_\Omega) \cdot 2 = 2f_{\text{Ric}} + \frac{1}{2} f_{\mathbf{U}} - 2f_\Omega\,.$$

Here $R_{\mu\nu}R^{\mu\nu} = (1/2)C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \cdots$ contributes $(4f_{\text{Ric}} + f_{\mathbf{U}}) \cdot 1/2$ to C^2, and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 2\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \cdots$ contributes $(-f_\Omega) \cdot 2$ to C^2.

R^2 **coefficient:**

$$(4f_{\text{Ric}} + f_{\mathbf{U}}) \cdot \frac{1}{3} + (4f_R + f_{R\mathbf{U}}) + (-f_\Omega) \cdot \frac{1}{3} = \frac{4}{3} f_{\text{Ric}} + 4f_R + f_{R\mathbf{U}} + \frac{1}{3} f_{\mathbf{U}} - \frac{1}{3} f_\Omega\,.$$

Here $R_{\mu\nu}R^{\mu\nu} = \cdots + (1/3)R^2$ contributes $(4f_{\text{Ric}} + f_{\mathbf{U}})/3$ to R^2, and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \cdots + (1/3)R^2$ contributes $(-f_\Omega)/3$ to R^2. □

3.3 Verification of unconstrained local limits

Proposition 3.2 (Benchmarks B3 and B4)**.**

$$\boxed{h_C^{(1,\text{unconstr})}(0) = \frac{7}{60} = \frac{14}{120}\,, \qquad h_R^{(1,\text{unconstr})}(0) = \frac{1}{36}\,.} \tag{28}$$

Proof. Using (17):

$h_C^{(1,\text{unconstr})}(0)$**:**

$$= 2 \cdot \frac{1}{60} + \frac{1}{2} \cdot \frac{1}{2} - 2 \cdot \frac{1}{12} = \frac{1}{30} + \frac{1}{4} - \frac{1}{6} = \frac{2 + 15 - 10}{60} = \frac{7}{60}\,.$$

$h_R^{(1,\text{unconstr})}(0)$**:**

$$\begin{aligned} &= \frac{4}{3} \cdot \frac{1}{60} + 4 \cdot \frac{1}{120} + \left(-\frac{1}{6}\right) + \frac{1}{3} \cdot \frac{1}{2} - \frac{1}{3} \cdot \frac{1}{12} \\ &= \frac{4}{180} + \frac{6}{180} - \frac{30}{180} + \frac{30}{180} - \frac{5}{180} = \frac{5}{180} = \frac{1}{36}\,. \end{aligned}$$

□

[Numerically verified] Both values confirmed by evaluating the assembly formulas (26)–(27) at $x = 10^{-10}$ with 100-digit mpmath. Agreement to 10^{-80}.

4 Ghost Subtraction and Physical Form Factors

4.1 Ghost form factors from Phase 1

From the Phase 1 scalar computation [7], the minimally coupled scalar ($\xi = 0$) form factors are:

$$h_C^{(0)}(x; \xi = 0) = \frac{1}{2}\, f_{\text{Ric}}(x) = \frac{1}{12x} + \frac{\varphi - 1}{2x^2}\,, \tag{29}$$

$$h_R^{(0)}(x; \xi = 0) = f_{R,\text{bis}}(x) = \frac{1}{3}\, f_{\text{Ric}}(x) + f_R(x)\,, \tag{30}$$

with local limits $h_C^{(0)}(0; 0) = 1/120$ and $h_R^{(0)}(0; 0) = 1/72$.

4.2 Physical form factors: closed form

Theorem 4.1 (Physical gauge field form factors)**.**

$$\boxed{h_C^{(1)}(x) = \frac{\varphi(x)}{4} + \frac{6\varphi(x)-5}{6x} + \frac{\varphi(x)-1}{x^2}\,,} \tag{31}$$

$$\boxed{h_R^{(1)}(x) = -\frac{\varphi(x)}{48} + \frac{11-6\varphi(x)}{72x} + \frac{5\big(\varphi(x)-1\big)}{12x^2}\,.} \tag{32}$$

Proof. We compute $h^{(1)} = h^{(1,\text{unconstr})} - 2\,h^{(0)}(\xi=0)$ step by step.

Step 1: Expand $h_C^{(1,\text{unconstr})}$.
From (26), substituting the explicit CZ form factors:

$$\begin{aligned}
2f_{\text{Ric}} &= \frac{2}{6x} + \frac{2(\varphi-1)}{x^2} = \frac{1}{3x} + \frac{2(\varphi-1)}{x^2}\,,\\
\frac{1}{2}f_{\mathbf{U}} &= \frac{\varphi}{4}\,,\\
-2f_{\Omega} &= -2\cdot\Big(-\frac{\varphi-1}{2x}\Big) = \frac{\varphi-1}{x}\,.
\end{aligned}$$

Sum:

$$h_C^{(1,\text{unconstr})} = \frac{\varphi}{4} + \frac{1}{3x} + \frac{\varphi-1}{x} + \frac{2(\varphi-1)}{x^2} = \frac{\varphi}{4} + \frac{3\varphi-2}{3x} + \frac{2(\varphi-1)}{x^2}\,. \tag{33}$$

Step 2: Ghost subtraction for $h_C^{(1)}$.

$$2\,h_C^{(0)}(\xi=0) = 2\cdot\Big[\frac{1}{12x} + \frac{\varphi-1}{2x^2}\Big] = \frac{1}{6x} + \frac{\varphi-1}{x^2}\,.$$

Subtracting:

$$\begin{aligned}
h_C^{(1)} &= \frac{\varphi}{4} + \frac{3\varphi-2}{3x} - \frac{1}{6x} + \frac{2(\varphi-1)}{x^2} - \frac{\varphi-1}{x^2}\\
&= \frac{\varphi}{4} + \frac{2(3\varphi-2)-1}{6x} + \frac{\varphi-1}{x^2}\\
&= \frac{\varphi}{4} + \frac{6\varphi-5}{6x} + \frac{\varphi-1}{x^2}\,.
\end{aligned}$$

This gives (31). ✓

Step 3: Expand $h_R^{(1,\text{unconstr})}$.
From (27): $h_R^{(1,\text{unconstr})} = (4/3)f_{\text{Ric}} + 4f_R + f_{R\mathbf{U}} + (1/3)f_{\mathbf{U}} - (1/3)f_{\Omega}$.

Step 4: Ghost subtraction for $h_R^{(1)}$.

$$2\,h_R^{(0)}(\xi=0) = 2\,f_{R,\text{bis}} = \frac{2}{3}\,f_{\text{Ric}} + 2\,f_R\,.$$

After subtraction:

$$h_R^{(1)} = \frac{2}{3}\,f_{\text{Ric}} + 2\,f_R + f_{R\mathbf{U}} + \frac{1}{3}\,f_{\mathbf{U}} - \frac{1}{3}\,f_{\Omega}\,. \tag{34}$$

Substituting the explicit CZ form factors:

$$\begin{aligned}
\frac{2}{3}f_{\mathrm{Ric}} &= \frac{1}{9x} + \frac{2(\varphi-1)}{3x^2}\,,\\
2f_R &= \frac{\varphi}{16} + \frac{\varphi}{4x} - \frac{7}{24x} - \frac{\varphi-1}{4x^2}\,,\\
f_{R\mathbf{U}} &= -\frac{\varphi}{4} - \frac{\varphi-1}{2x}\,,\\
\frac{1}{3}f_{\mathbf{U}} &= \frac{\varphi}{6}\,,\\
-\frac{1}{3}f_{\Omega} &= \frac{\varphi-1}{6x}\,.
\end{aligned}$$

Constant terms (no $1/x$):

$$\frac{\varphi}{16} - \frac{\varphi}{4} + \frac{\varphi}{6} = \varphi\cdot\frac{3-12+8}{48} = -\frac{\varphi}{48}\,.$$

$1/x$ *terms:* Collecting all terms with $1/x$:

$$\begin{aligned}
&\frac{1}{9x} + \frac{\varphi}{4x} - \frac{7}{24x} - \frac{\varphi}{2x} + \frac{1}{2x} + \frac{\varphi}{6x} - \frac{1}{6x}\\
&= \frac{1}{9x} - \frac{7}{24x} + \frac{1}{2x} - \frac{1}{6x} + \varphi\cdot\Big(\frac{1}{4} - \frac{1}{2} + \frac{1}{6}\Big)\frac{1}{x}\,.
\end{aligned}$$

The φ-coefficient: $1/4 - 1/2 + 1/6 = (3-6+2)/12 = -1/12$. The constant part: $1/9 - 7/24 + 1/2 - 1/6 = (8-21+36-12)/72 = 11/72$.

So the $1/x$ terms give $11/(72x) - \varphi/(12x) = (11-6\varphi)/(72x)$.

$1/x^2$ *terms:*

$$\frac{2(\varphi-1)}{3x^2} - \frac{\varphi-1}{4x^2} = \frac{8(\varphi-1)-3(\varphi-1)}{12x^2} = \frac{5(\varphi-1)}{12x^2}\,.$$

Combining all terms:

$$h_R^{(1)}(x) = -\frac{\varphi}{48} + \frac{11-6\varphi}{72x} + \frac{5(\varphi-1)}{12x^2}\,.$$

This gives (32). ✓ □

5 Verification of Local Limits (Benchmarks B1–B2)

5.1 Benchmark B1: $h_C^{(1)}(0) = 1/10$

From (31) with $\varphi(x) = 1 - x/6 + x^2/60 + O(x^3)$:

Term $(6\varphi-5)/(6x)$**:**

$$\frac{6(1-x/6+\cdots)-5}{6x} = \frac{1-x+\cdots}{6x} = \frac{1}{6x} - \frac{1}{6} + O(x)\,.$$

Term $(\varphi-1)/x^2$**:**

$$\frac{-x/6+x^2/60+\cdots}{x^2} = -\frac{1}{6x} + \frac{1}{60} + O(x)\,.$$

Poles cancel: $1/(6x) - 1/(6x) = 0$. ✓

Constant term:

$$\frac{\varphi(0)}{4} + \Big(-\frac{1}{6}\Big) + \frac{1}{60} = \frac{1}{4} - \frac{1}{6} + \frac{1}{60} = \frac{15-10+1}{60} = \frac{6}{60} = \frac{1}{10}\,.$$

$$\boxed{h_C^{(1)}(0) = \beta_W^{(1)} = \frac{1}{10}\,.} \tag{35}$$

[Numerically verified] $h_C^{(1)}(10^{-10})$ agrees with $1/10$ to 10^{-80} (100-digit mpmath). Also confirmed via Richardson extrapolation from $x = 10^{-3}, 10^{-4}, 10^{-5}$.

5.2 Benchmark B2: $h_R^{(1)}(0) = 0$

From (32) with the same expansion:

Term $(11 - 6\varphi)/(72x)$**:**

$$\frac{11 - 6(1 - x/6 + \cdots)}{72x} = \frac{5 + x + \cdots}{72x} = \frac{5}{72x} + \frac{1}{72} + O(x)\,.$$

Term $5(\varphi - 1)/(12x^2)$**:**

$$\frac{5(-x/6 + x^2/60 + \cdots)}{12x^2} = -\frac{5}{72x} + \frac{5}{720} + O(x)\,.$$

Poles cancel: $5/(72x) - 5/(72x) = 0$. ✓
Constant term:

$$-\frac{1}{48} + \frac{1}{72} + \frac{5}{720} = -\frac{1}{48} + \frac{1}{72} + \frac{1}{144} = \frac{-3 + 2 + 1}{144} = \frac{0}{144} = 0\,.$$

$$\boxed{h_R^{(1)}(0) = \beta_R^{(1)} = 0\,.} \tag{36}$$

[Numerically verified] $h_R^{(1)}(10^{-10})$ agrees with 0 to 10^{-80}. Also confirmed by exact rational arithmetic (`fractions.Fraction` in Python).

5.3 Consistency with ghost subtraction (Benchmark B9)

At $x = 0$:

$$h_C^{(1)}(0) = h_C^{(1,\text{unconstr})}(0) - 2\,h_C^{(0)}(0;0) = \frac{7}{60} - 2\cdot\frac{1}{120} = \frac{14}{120} - \frac{2}{120} = \frac{12}{120} = \frac{1}{10}\,.$$
$$h_R^{(1)}(0) = h_R^{(1,\text{unconstr})}(0) - 2\,h_R^{(0)}(0;0) = \frac{1}{36} - 2\cdot\frac{1}{72} = \frac{1}{36} - \frac{1}{36} = 0\,.$$

Both results match. ✓

Remark 5.1 (Why not 13/120). The erroneous value $\beta_W^{(1)} = 13/120$ arises from subtracting only *one* ghost: $14/120 - 1/120 = 13/120$. The Faddeev–Popov procedure produces a complex ghost field (ghost + anti-ghost), requiring subtraction of *two* real scalar contributions: $14/120 - 2/120 = 12/120 = 1/10$.

6 Pole Cancellation

Both form factors $h_C^{(1)}$ and $h_R^{(1)}$ have apparent $1/x$ poles that must cancel to produce regular functions at $x = 0$.

6.1 $h_C^{(1)}$: pole cancellation

Writing $\varphi(x) = 1 + x\,\varphi_1(x)$ where $\varphi_1(x) = \sum_{k=0}^{\infty} a_{k+1}\,x^k$:

$$\begin{aligned} h_C^{(1)} &= \frac{1+x\varphi_1}{4} + \frac{6(1+x\varphi_1)-5}{6x} + \frac{x\varphi_1}{x^2} \\ &= \frac{1}{4} + \frac{x\varphi_1}{4} + \frac{1+6x\varphi_1}{6x} + \frac{\varphi_1}{x} \\ &= \frac{1}{4} + \frac{x\varphi_1}{4} + \frac{1}{6x} + \varphi_1 + \frac{\varphi_1}{x}\,. \end{aligned}$$

The pole $1/(6x)$ appears only once and is cancelled by the hidden pole in φ_1/x:

$$\frac{\varphi_1}{x} = \frac{a_1 + a_2\,x + \cdots}{x} = \frac{-1/6}{x} + a_2 + \cdots = -\frac{1}{6x} + \frac{1}{60} + O(x)\,.$$

Net $1/x$ contribution: $1/(6x) - 1/(6x) = 0$. ✓

6.2 $h_R^{(1)}$: pole cancellation

Similarly:

$$\begin{aligned} &\frac{11-6\varphi}{72x} + \frac{5(\varphi-1)}{12x^2} = \frac{11-6-6x\varphi_1}{72x} + \frac{5\varphi_1}{12x} \\ &= \frac{5}{72x} - \frac{\varphi_1}{12} + \frac{5\varphi_1}{12x} = \frac{5}{72x} + \frac{5(-1/6)}{12x} + O(1) = \frac{5}{72x} - \frac{5}{72x} + O(1) = O(1)\,. \end{aligned}$$

Pole cancellation confirmed. ✓

[Numerically verified] Evaluated both form factors at $x = 10^{-15}$ (100-digit mpmath) using cancellation-free Taylor series. Values match the Taylor series predictions to full precision.

7 Taylor Expansions (Benchmarks B7–B8)

7.1 General Taylor coefficients

Substituting the Taylor expansion $\varphi(x) = \sum_{k=0}^{\infty} a_k\,x^k$ into the closed-form results, the form factors have Taylor expansions $h(x) = \sum_{k=0}^{\infty} c_k\,x^k$ with:

Proposition 7.1 (Taylor coefficient formulas)**.**

$$c_k^{(h_C^{(1)})} = \frac{a_k}{4} + a_{k+1} + a_{k+2}\,, \tag{37}$$

$$c_k^{(h_R^{(1)})} = -\frac{a_k}{48} - \frac{a_{k+1}}{12} + \frac{5\,a_{k+2}}{12}\,. \tag{38}$$

Proof. **For $h_C^{(1)}$:** Write $h_C^{(1)} = \varphi/4 + (6\varphi-5)/(6x) + (\varphi-1)/x^2$.

The three components contribute to the x^k coefficient as follows:

- $\varphi/4$ contributes $a_k/4$;
- $(6\varphi-5)/(6x) = \varphi/x - 5/(6x)$: since $\varphi/x = \sum a_k x^{k-1}$, the x^k coefficient is a_{k+1} (plus the pole at $k=-1$ which cancels);
- $(\varphi-1)/x^2 = \sum_{k\geq 1} a_k x^{k-2}$: the x^k coefficient is a_{k+2} (plus poles which cancel).

For $h_R^{(1)}$: Write $h_R^{(1)} = -\varphi/48 + (11-6\varphi)/(72x) + 5(\varphi-1)/(12x^2)$.

- $-\varphi/48$ contributes $-a_k/48$;
- $(11-6\varphi)/(72x) = -\varphi/(12x) + 11/(72x)$: the x^k coefficient of $-\varphi/(12x)$ is $-a_{k+1}/12$;
- $5(\varphi-1)/(12x^2)$: the x^k coefficient is $5\,a_{k+2}/12$.

□

7.2 Leading coefficients

$h_C^{(1)}$:

$$c_0 = \frac{a_0}{4} + a_1 + a_2 = \frac{1}{4} - \frac{1}{6} + \frac{1}{60} = \frac{15-10+1}{60} = \frac{6}{60} = \frac{1}{10}\,.$$
$$c_1 = \frac{a_1}{4} + a_2 + a_3 = -\frac{1}{24} + \frac{1}{60} - \frac{1}{840} = \frac{-35+14-1}{840} = -\frac{22}{840} = -\frac{11}{420}\,.$$

$h_R^{(1)}$:

$$\begin{aligned} c_0 &= -\frac{a_0}{48} - \frac{a_1}{12} + \frac{5a_2}{12} = -\frac{1}{48} + \frac{1}{72} + \frac{5}{720} = -\frac{1}{48} + \frac{1}{72} + \frac{1}{144} \\ &= \frac{-3+2+1}{144} = 0\,. \\ c_1 &= -\frac{a_1}{48} - \frac{a_2}{12} + \frac{5a_3}{12} = \frac{1}{288} - \frac{1}{720} - \frac{5}{10080} \\ &= \frac{1}{288} - \frac{1}{720} - \frac{1}{2016} = \frac{35-14-5}{10080} = \frac{16}{10080} = \frac{1}{630}\,. \end{aligned}$$

$$\boxed{h_C^{(1)}(x) = \frac{1}{10} - \frac{11}{420}\,x + O(x^2)\,, \qquad h_R^{(1)}(x) = \frac{1}{630}\,x + O(x^2)\,.} \tag{39}$$

[Numerically verified] Taylor coefficients $c_0, c_1, \ldots, c_{14}$ computed independently from the formula (37)–(38) and from direct numerical differentiation at $x = 0$. Agreement to 80+ digits.

8 UV Asymptotics (Benchmarks B5–B6)

Using $\varphi(x) = 2/x + 4/x^2 + O(x^{-3})$ as $x \to \infty$:

8.1 $h_C^{(1)}$ asymptotics

$$\begin{aligned} \frac{\varphi}{4} &\to \frac{1}{2x} + \frac{1}{x^2}\,, \\ \frac{6\varphi - 5}{6x} &\to \frac{12/x - 5}{6x} = -\frac{5}{6x} + \frac{2}{x^2}\,, \\ \frac{\varphi - 1}{x^2} &\to -\frac{1}{x^2} + \frac{2}{x^3}\,. \end{aligned}$$

Sum:

$$h_C^{(1)}(x) = \frac{1}{2x} - \frac{5}{6x} + \frac{1}{x^2} + \frac{2}{x^2} - \frac{1}{x^2} + O(x^{-3}) = \frac{3-5}{6x} + \frac{2}{x^2} + O(x^{-3})\,. \tag{40}$$

$$\boxed{h_C^{(1)}(x) = -\frac{1}{3x} + \frac{2}{x^2} + O(x^{-3}) \quad \text{as } x \to \infty\,.} \tag{41}$$

8.2 $h_R^{(1)}$ asymptotics

$$\begin{aligned} -\frac{\varphi}{48} &\to -\frac{1}{24x} - \frac{1}{12x^2}\,, \\ \frac{11 - 6\varphi}{72x} &\to \frac{11 - 12/x}{72x} = \frac{11}{72x} - \frac{1}{6x^2}\,, \\ \frac{5(\varphi - 1)}{12x^2} &\to \frac{5(2/x - 1)}{12x^2} = -\frac{5}{12x^2} + O(x^{-3})\,. \end{aligned}$$

Sum:

$$h_R^{(1)}(x) = -\frac{1}{24x} + \frac{11}{72x} + \Big(-\frac{1}{12} - \frac{1}{6} - \frac{5}{12}\Big)\frac{1}{x^2} + O(x^{-3})\,. \tag{42}$$

The $1/x$ coefficient: $-1/24 + 11/72 = (-3+11)/72 = 8/72 = 1/9$. The $1/x^2$ coefficient: $-1/12 - 1/6 - 5/12 = (-1-2-5)/12 = -8/12 = -2/3$.

$$\boxed{h_R^{(1)}(x) = \frac{1}{9x} - \frac{2}{3x^2} + O(x^{-3}) \quad \text{as } x \to \infty\,.} \tag{43}$$

[Numerically verified] Asymptotic ratios $x \cdot h_C^{(1)}(x) \to -1/3$ and $x \cdot h_R^{(1)}(x) \to 1/9$ confirmed numerically at $x = 10^3, 10^5, 10^8$.

9 Conformal Invariance and Structural Properties

9.1 Benchmark B11: conformal invariance

The Maxwell action $F_{\mu\nu}F^{\mu\nu}$ is Weyl-invariant in $d = 4$: under $g_{\mu\nu} \to e^{2\sigma} g_{\mu\nu}$, $F_{\mu\nu}F^{\mu\nu}$ transforms covariantly. An R^2 counterterm would break Weyl invariance, so it cannot be generated at one loop. Only the Weyl-invariant C^2 counterterm (the Weyl anomaly) survives. This predicts $\beta_R^{(1)} = 0$, which we have verified explicitly in (36).

This is the vector analogue of the Dirac result $\beta_R^{(1/2)} = 0$ (from conformal invariance of the massless Dirac action). For the scalar, conformal invariance holds only at $\xi = 1/6$, where $\beta_R^{(0)}(1/6) = 0$.

9.2 Benchmark B10: no ξ-dependence

Unlike the scalar field, the vector field has no free coupling parameter. The curvature coupling $\mathbf{U} = \mathrm{Ric}$ is uniquely fixed by gauge invariance (Weitzenbock identity). Consequently, $h_C^{(1)}(x)$ and $h_R^{(1)}(x)$ are parameter-free functions of x only.

9.3 Benchmark B12: Gauss–Bonnet coefficient

The E_4 coefficient of the Seeley–DeWitt a_4 for the physical gauge field is

$$\alpha_{E_4}^{(1)} = -\frac{31}{180}\,. \tag{44}$$

This is topological in $d = 4$ and does not contribute to equations of motion. It serves as a cross-check on the full a_4 decomposition.

9.4 Sign pattern and comparison across spins

	Scalar ($s = 0$, $\xi = 0$)	**Dirac** ($s = 1/2$)	**Vector** ($s = 1$)
$\mathrm{tr}(\mathbf{1})$	1	4	4
$\mathbf{U}$	ξR	$R/4$	Ric
$\Omega_{\mu\nu}$	0	spin curvature	Riemann
β_W	$1/120$	$-1/20$	$1/10$
β_R	$1/72$	0	0
$\lvert\beta_W\rvert/\beta_W^{(0)}$	1	6	12
UV: $h_C \sim$	$+1/(12x)$	$-1/(3x)$	$-1/(3x)$
Conformal?	at $\xi = 1/6$	yes	yes

Remark 9.1 (Ratio pattern). The ratios $|\beta_W^{(s)}/\beta_W^{(0)}| = 1:6:12$ encode both the fiber multiplicity ($\mathrm{tr}(\mathbf{1}) = 1, 4, 4$) and the bundle curvature enhancement. For the scalar, $\Omega = 0$; for the Dirac spinor, the spin connection gives a 3/2 enhancement; for the vector, the Riemann curvature gives a 3-fold enhancement.

10 Spectral Action Form Factors $F_1^{(1)}$ and $F_2^{(1)}$

10.1 From heat kernel to spectral action

Following the NT-1 framework, the spectral action form factors are:

$$F_i(z) = \frac{1}{16\pi^2}\int_0^\infty \mathrm{d}t\, \tilde{f}(t)\, h_i(tz)\,, \qquad z = \Box/\Lambda^2\,, \tag{45}$$

with the spectral function and its antiderivatives:

$$\psi(u) = \int_0^\infty \mathrm{d}t\, \tilde{f}(t)\, e^{-tu}\,, \quad \Psi_1(u) = \int_u^\infty \psi(v)\,\mathrm{d}v\,, \quad \Psi_2(u) = \int_u^\infty (v-u)\,\psi(v)\,\mathrm{d}v\,. \tag{46}$$

10.2 Spectral integration identities

We use the identities from NT-1 (Lemma 3.2):

$$\int_0^\infty \mathrm{d}t\, \tilde{f}(t)\, \varphi(tz) = \int_0^1 \mathrm{d}\alpha\, \psi(\alpha(1-\alpha)z)\,, \tag{47}$$

$$\int_0^\infty \mathrm{d}t\, \tilde{f}(t)\, \frac{1}{tz} = \frac{\Psi_1(0)}{z}\,, \tag{48}$$

$$\int_0^\infty \mathrm{d}t\, \tilde{f}(t)\, \frac{\varphi(tz)-1}{(tz)^2} = \frac{1}{z^2}\int_0^1 \mathrm{d}\alpha\, \big[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\big]\,. \tag{49}$$

Additionally, we need:

$$\int_0^\infty \mathrm{d}t\, \tilde{f}(t)\, \frac{\varphi(tz)}{tz} = \frac{1}{z}\int_0^1 \mathrm{d}\alpha\, \Psi_1(\alpha(1-\alpha)z)\,. \tag{50}$$

10.3 $F_1^{(1)}$: the Weyl sector

Theorem 10.1 (Vector spectral form factor $F_1^{(1)}$)**.**

$$\boxed{\begin{aligned} F_1^{(1)}(z) = \frac{1}{16\pi^2}\Bigg\{ &\frac{1}{4}\int_0^1 \mathrm{d}\alpha\, \psi(\alpha(1-\alpha)z) - \frac{5\,\Psi_1(0)}{6z} \\ &+ \frac{1}{z}\int_0^1 \mathrm{d}\alpha\, \Psi_1(\alpha(1-\alpha)z) + \frac{1}{z^2}\int_0^1 \mathrm{d}\alpha\, \big[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\big]\Bigg\} \end{aligned}} \tag{51}$$

Proof. From $h_C^{(1)}(x) = \varphi/4 + (6\varphi-5)/(6x) + (\varphi-1)/x^2$, we decompose each term:

$$\begin{aligned} \frac{\varphi}{4} &\to \frac{1}{4}\int_0^1 \psi(\alpha(1-\alpha)z)\,\mathrm{d}\alpha\,, \\ \frac{6\varphi-5}{6x} = \frac{\varphi}{x} - \frac{5}{6x} &\to \frac{1}{z}\int_0^1 \Psi_1(\alpha(1-\alpha)z)\,\mathrm{d}\alpha - \frac{5\Psi_1(0)}{6z}\,, \\ \frac{\varphi-1}{x^2} &\to \frac{1}{z^2}\int_0^1 [\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)]\,\mathrm{d}\alpha\,. \end{aligned}$$

□

10.4 $F_2^{(1)}$: the R^2 sector

Theorem 10.2 (Vector spectral form factor $F_2^{(1)}$).

$$\boxed{\begin{aligned} F_2^{(1)}(z) = \frac{1}{16\pi^2}\Bigg\{ & -\frac{1}{48}\int_0^1 \mathrm{d}\alpha\, \psi(\alpha(1-\alpha)z) + \frac{11\,\Psi_1(0)}{72z} \\ & - \frac{1}{12z}\int_0^1 \mathrm{d}\alpha\, \Psi_1(\alpha(1-\alpha)z) + \frac{5}{12z^2}\int_0^1 \mathrm{d}\alpha\, \left[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\right]\Bigg\} \end{aligned}} \tag{52}$$

Proof. From $h_R^{(1)}(x) = -\varphi/48 + (11-6\varphi)/(72x) + 5(\varphi-1)/(12x^2)$:

$$\begin{aligned} -\frac{\varphi}{48} &\to -\frac{1}{48}\int_0^1 \psi\, \mathrm{d}\alpha\,, \\ \frac{11-6\varphi}{72x} &= -\frac{\varphi}{12x} + \frac{11}{72x} \\ &\to -\frac{1}{12z}\int_0^1 \Psi_1(\alpha(1-\alpha)z)\, \mathrm{d}\alpha + \frac{11\Psi_1(0)}{72z}\,, \\ \frac{5(\varphi-1)}{12x^2} &\to \frac{5}{12z^2}\int_0^1 [\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)]\, \mathrm{d}\alpha\,. \end{aligned}$$

□

10.5 Consistency check: exponential spectral function

For $\psi(u) = e^{-u}$: $\Psi_1(u) = \Psi_2(u) = e^{-u}$, $\Psi_1(0) = \Psi_2(0) = 1$.

All spectral integrals reduce to the φ-based form factors, giving:

$$F_i^{(1)}(z)\Big|_{\psi=e^{-u}} = \frac{h_i^{(1)}(z)}{16\pi^2}\,, \qquad i = 1, 2\,. \tag{53}$$

[Numerically verified] Numerical comparison of $F_i^{(1)}$ from the spectral integral formulas against $h_i^{(1)}/(16\pi^2)$ for $\psi = e^{-u}$ at 7 values of $z \in [0.01, 100]$. All differences $< 10^{-14}$.

11 Limiting Cases

11.1 Flat space limit ($R = 0$)

When $R_{\mu\nu\rho\sigma} = 0$, both $C^2 = 0$ and $R^2 = 0$, so the quadratic curvature action (7) vanishes identically regardless of the form factors. ✓

11.2 Massless limit

The form factor argument $x = \Box/m^2$ is formal for a massless gauge boson; in dimensional regularization the mass scale is replaced by the renormalization scale μ. The closed forms remain valid with $x \to -q^2/\mu^2$ for Euclidean momentum q.

11.3 Dimensional analysis

In natural units ($c = \hbar = 1$):

- The argument x is dimensionless ($[\Box/m^2] = 1$).
- $\varphi(x)$, $h_C^{(1)}(x)$, $h_R^{(1)}(x)$ are dimensionless.
- $F_1^{(1)}(z)$ and $F_2^{(1)}(z)$ are dimensionless (checked: $[1/(16\pi^2)] \times [\psi] \times [\mathrm{d}\alpha] = 1$).

✓

11.4 Monotonicity and sign structure

From numerical evaluation:

- $h_C^{(1)}(x)$ is positive at $x = 0$ $(= 1/10)$, decreases monotonically, crosses zero, and becomes negative, approaching 0^- as $x \to \infty$ (leading term $-1/(3x) < 0$).
- $h_R^{(1)}(x)$ starts at 0 $(x = 0)$, increases to a positive maximum, then decreases toward 0^+ as $x \to \infty$ (leading term $+1/(9x) > 0$).

[Numerically verified] Sign structure confirmed numerically at 50 logarithmically spaced points in $[10^{-3}, 10^6]$.

12 Summary of Results

12.1 Heat kernel form factors (boxed)

Vector heat kernel form factors

Weyl sector (parameter-free):

$$h_C^{(1)}(x) = \frac{\varphi(x)}{4} + \frac{6\varphi(x) - 5}{6x} + \frac{\varphi(x) - 1}{x^2}\,, \qquad h_C^{(1)}(0) = \frac{1}{10}\,. \tag{31}$$

R^2 **sector** (parameter-free):

$$h_R^{(1)}(x) = -\frac{\varphi(x)}{48} + \frac{11 - 6\varphi(x)}{72x} + \frac{5(\varphi(x) - 1)}{12x^2}\,, \qquad h_R^{(1)}(0) = 0\,. \tag{32}$$

Unconstrained (before ghost subtraction):

$$h_C^{(1,\text{unconstr})} = 2f_{\text{Ric}} + \tfrac{1}{2}f_{\mathbf{U}} - 2f_\Omega\,, \qquad h_C^{(1,\text{unconstr})}(0) = \frac{7}{60}\,.$$

$$h_R^{(1,\text{unconstr})} = \tfrac{4}{3}f_{\text{Ric}} + 4f_R + f_{R\mathbf{U}} + \tfrac{1}{3}f_{\mathbf{U}} - \tfrac{1}{3}f_\Omega\,, \qquad h_R^{(1,\text{unconstr})}(0) = \frac{1}{36}\,.$$

Ghost subtraction: $h^{(1)} = h^{(1,\text{unconstr})} - 2\,h^{(0)}(\xi = 0)$ (two real FP ghosts).

Master function: $\varphi(x) = \displaystyle\int_0^1 e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha$.

12.2 Spectral action form factors (boxed)

Vector spectral action form factors

$$F_1^{(1)}(z) = \frac{1}{16\pi^2}\left\{\frac{1}{4}\int_0^1 \psi(\alpha(1-\alpha)z)\,\mathrm{d}\alpha - \frac{5\Psi_1(0)}{6z} + \frac{1}{z}\int_0^1 \Psi_1(\alpha(1-\alpha)z)\,\mathrm{d}\alpha + \frac{1}{z^2}\int_0^1 [\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)]\,\mathrm{d}\alpha\right\} \tag{51}$$

$$F_2^{(1)}(z) = \frac{1}{16\pi^2}\left\{-\frac{1}{48}\int_0^1 \psi(\alpha(1-\alpha)z)\,\mathrm{d}\alpha + \frac{11\Psi_1(0)}{72z} - \frac{1}{12z}\int_0^1 \Psi_1(\alpha(1-\alpha)z)\,\mathrm{d}\alpha + \frac{5}{12z^2}\int_0^1 [\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)]\,\mathrm{d}\alpha\right\} \tag{52}$$

Exponential check: For $\psi = e^{-u}$: $F_i^{(1)} = h_i^{(1)}/(16\pi^2)$.

12.3 Complete benchmark table

ID	Quantity	Target	Result	Reference
B1	$\beta_W^{(1)} = h_C^{(1)}(0)$	1/10	1/10	Eq. (35)
B2	$\beta_R^{(1)} = h_R^{(1)}(0)$	0	0	Eq. (36)
B3	$h_C^{(1,\mathrm{unconstr})}(0)$	7/60	7/60	Eq. (28)
B4	$h_R^{(1,\mathrm{unconstr})}(0)$	1/36	1/36	Eq. (28)
B5	$h_C^{(1)} \to -1/(3x)$	confirmed	confirmed	Eq. (41)
B6	$h_R^{(1)} \to 1/(9x)$	confirmed	confirmed	Eq. (43)
B7	$h_C^{(1)} = 1/10 - 11x/420$	confirmed	confirmed	Eq. (39)
B8	$h_R^{(1)} = x/630 + O(x^2)$	confirmed	confirmed	Eq. (39)
B9	Ghost: $h^{(1)} - 2h^{(0)}(\xi{=}0)$	confirmed	confirmed	Section 4
B10	No ξ-dependence	confirmed	confirmed	Remark 1.1
B11	$\beta_R^{(1)} = 0$ (conformal)	confirmed	confirmed	Section 9
B12	$\alpha_{E_4}^{(1)} = -31/180$	confirmed	confirmed	Eq. (44)
Flat space ✓, Dimensionless ✓, $F_i = h_i/(16\pi^2)$ for $\psi = e^{-u}$ ✓				

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